Summary

The types of straight lines related to the information.

S.No	Information given	Equation of the lines
1	Slope(m) and y-intercept (b)	$y = mx + b$
2	Slope (m) and point (x_1, y_1)	$y - y_1 = m(x - x_1)$
3	Two points(x_1, y_1) and (x_2, y_2)	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$
4	x-intercept (a) and y-intercept (b)	$\frac{x}{a} + \frac{y}{b} = 1$
5	Normal length (p) , angle (α)	$x \cos\alpha + y \sin\alpha = p$
6	Parametric form: parameter-r	$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$
7	The general equation	$ax + by + c = 0$

Form of lines	Condition for parallel	Condition for perpendicular
$y = m_1x + c_1$ and $y = m_2x + c_2$	$m_2 = m_1$	$m_1m_2 = -1$
$a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$	$a_1b_2 = a_2b_1$	$a_1a_2 + b_1b_2 = 0$

A point $P(x_1, y_1)$is on the origin side or non origin side of the line $ax + by + c = 0$ $(c \neq 0)$, according as $ax_1 + by_1 + c$ and c are of the same sign or opposite sign.

The distance between two points (x_1, y_1) and (x_2, y_2) is given by the formula

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The distance from a point $P(x_1, y_1)$ to a line $ax + by + c = 0$ is $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$

The distance between two parallel lines $a_1x + b_1y + c_1 = 0$ and $a_1x + b_1y + c_2 = 0$ is $\frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$

The line parallel to $ax + by + c = 0$ through a point $(x_1,\ y_1)$, is $ax + by = ax_1 + by_1$ and the perpendicular line is $bx - ay = bx_1 - ay_1$